Day 11: Probability Basics
Understanding Uncertainty in Data Science and Machine Learning

100 Days of Data Science

Why Probability is the Heart of Data Science

Real-world data is never perfect or certain. Machine Learning models do not give exact answers — they give probabilities.

Probability helps us:

- Measure uncertainty
- Make predictions
- Evaluate risk
- Build intelligent systems

Almost every ML model internally works using probability.

What is Probability?

Probability is a numerical value that measures how likely an event is to occur.

Its value always lies between:

$$0 \leq P \leq 1$$

Where:

- 0 means impossible
- 1 means certain

Basic Probability Formula

$$P(Event) = \frac{\text{Number of favorable outcomes}}{\text{Total possible outcomes}}$$

Simple Real-Life Example

Tossing a coin:

Outcomes = Head, Tail

Probability of Head:

$$P(Head) = \frac{1}{2}$$

Dice Example

Rolling a dice:

Total outcomes = 6

Probability of getting even number (2,4,6):

$$P(Even) = \frac{3}{6} = 0.5$$

Real-World Example: Weather

If weather app says:

Probability of rain = 0.8

It means:

- Very likely to rain
- But not guaranteed

Types of Events

Certain Event

An event that will always happen.

Example: Sun rises tomorrow.

Impossible Event

An event that cannot happen.

Example: Getting 10 on a dice.

Random Event

An event that may or may not happen.

Example: Stock market going up tomorrow.

Complement of an Event

If event A happens with probability:

$$P(A)$$

Then probability of NOT A:

$$P(\text{Not A}) = 1 - P(A)$$

Example:

If probability of passing exam = 0.9

Probability of failing = 0.1

Probability in Machine Learning (Very Important)

Classification Models

ML models like Logistic Regression, Naive Bayes, Neural Networks give output as:

- Probability of each class

Example:

Email spam model:

- Spam = 0.95
- Not spam = 0.05

Model predicts spam because probability is higher.

Confidence in Predictions

Higher probability = higher confidence
Lower probability = uncertain prediction
This helps in:

- Medical diagnosis
- Fraud detection
- Autonomous vehicles

Risk and Probability

Banks use probability to calculate:

- Loan default chance
- Credit score risk

Insurance uses probability to:

- Price policies
- Predict claims

Common Probability Mistakes

- Thinking high probability means certainty
- Ignoring randomness
- Confusing chance with guarantee

Why Probability Makes ML Powerful

Machine Learning does not say:

"This is spam."

It says:

"There is 95% chance this is spam."

This allows:

- Better decisions
- Threshold tuning
- Risk control

Final Takeaway

Probability is the language of uncertainty.

Without probability:

- Machine Learning would not exist
- Predictions would be guesses

Understanding probability is essential to becoming a Data Scientist.